Tommaso Aiello

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EDUCATION

Bocconi University **Milan, Italy | Sep 2025 – Jul 2027**
MSc in Artificial Intelligence | Current average: 28.6/30
- Associazione San Paolo merit scholar (1 of 20; full tuition).
- University of Maryland exchange, Jan–Jun 2027.

Bocconi University **Milan, Italy | Sep 2022 – Jul 2025**
BSc in Applied Mathematics for Artificial Intelligence | Final grade: 108/110
- Thesis: Diffusion Processes for Generative Hierarchical Models.
- City University of Hong Kong exchange, Jan–Jun 2025.
- Coursework: Mathematical Statistics (30 cum laude), Mathematical Modelling for Finance (30), Stochastic Processes (29), Optimisation and Inference (30 cum laude).

PROFESSIONAL EXPERIENCE

BNL BNP Paribas — Data Analyst Intern **Rome, Italy | Jul–Sep 2024**
- Built a SQL/Python/Excel pipeline analysing over 10,000 positions and nearly 300,000 transactions across 397 portfolios.
- Forecast year-end Net Promoter Score (NPS) within 0.3 points using short-horizon Monte Carlo; validated by MSE and cross-entropy.
- Deployed a Power BI/Excel dashboard for volumes, client returns, benchmark performance and allocation; adopted permanently after testing.

QUANTITATIVE RESEARCH & SELECTED PROJECTS

BAINSA — Cryptocurrency Market Trend Forecasting **Mar–Sep 2024**
- Analysed six months of hourly data for ~100 cryptocurrencies and ~1m X/Reddit posts using FinBERT, correlation and OLS; retained significance after Holm/FDR correction.
- Built a sentiment-adjusted Black–Scholes implied-volatility framework; a 30m-parameter network predicted its correction term (R^2 0.78).

Bocconi University — Master's Research with Marc Mézard **Jan 2026–Present**
- Derived closed-form likelihoods, denoising scores and spinodal times for Curie–Weiss and hierarchical mean-field models under Ornstein–Uhlenbeck diffusion.
- Validated analytical phase boundaries on 130,000 Monte Carlo samples, with pooled RMSE 0.038 and 95%-crossing errors of 0.019–0.027.
- Extending the analytical and Monte Carlo results toward a master's thesis and potential publication.

Bachelor's Thesis — Diffusion Processes for Generative Hierarchical Models **2025**
- Built a reproducible Transformer denoiser; ran 1,000+ experiments and earned the highest thesis mark.
- Found phase transitions required sufficient hierarchy and dimensionality; low-structure regimes memorised or failed to learn.

Pydeia — 1st of 24 teams, InnovAIte Hackathon **Nov 2025**
- Developed the data pipeline and multimodal recommender for 103 Italian universities, combining three semantic embeddings (academic, aspirations and lifestyle) with budget and geography constraints to generate explainable top-three recommendations.

SKILLS & ADDITIONAL INFORMATION

- **Programming:** Python (PyTorch, NumPy, pandas, scikit-learn), SQL, R, C/C++, MATLAB, JavaScript, Git.
- **Methods & tools:** Time-series modelling, financial NLP, Monte Carlo, stochastic processes, Markov chains, OLS, optimisation, model validation, Excel, Power BI.
- **Languages:** Italian (native), English (Cambridge C1; IELTS 7.5), Spanish (conversational).
- **Additional:** Fully funded Ottawa exchange, Jan–Jul 2020; competitive swimming; bass guitar.